medimg

Compressed Sensing Reconstruction for Accelerated MRI and CT

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Abstract

medimg is a research-grade Python package for compressed sensing and model-based image reconstruction in accelerated magnetic resonance imaging (MRI) and sparse-view computed tomography (CT). It implements three classical variational methods — Tikhonov regularisation (conjugate gradient), isotropic total variation (Chambolle-Pock primal-dual), and wavelet-domain ℓ_1 minimisation (FISTA) — with clean operator-based abstractions that work across both modalities. Variable-density Cartesian k-space under-sampling (Lustig et al., 2007) and sparse-angle parallel-beam Radon projections (Sidky & Pan, 2008) serve as the forward models. The package includes two demonstration scripts that generate publication-quality comparison figures, and a suite of 41 unit tests.

1 Overview

Accelerated acquisition in MRI and CT reduces scan time and radiation dose, but the resulting under-sampled data lead to an ill-posed linear inverse problem. In MRI, k-space is sub-sampled below the Nyquist rate; in CT, the number of projection angles is reduced. Compressed sensing (CS) theory [3, 4] guarantees exact recovery under sparsity and incoherence assumptions, provided a suitable nonlinear reconstruction algorithm is employed.

This technical note describes the forward models (Section 2), the compressed sensing principles that underpin the reconstruction (Section 3), the three variational methods implemented in the package (Section 4), and the software structure and usage (Section 5).

2 Forward Models

2.1 MRI

The single-coil MRI measurement process is the 2-D Fourier transform sampled at discrete k-space locations. Let $\mathcal{F}$ denote the unitary 2-D DFT, $M \in \{0,1\}^{n_r \times n_c}$ a binary under-sampling mask, and $\eta \sim \mathcal{CN}(0, \sigma^2 I)$ additive complex Gaussian noise. The forward operator is

$$A_{\mathrm{MRI}}(x) = M \odot \mathcal{F}(x), \qquad y = A_{\mathrm{MRI}}(x) + \eta, \tag{1}$$

with adjoint $A^*_{\mathrm{MRI}}(y) = \mathrm{Re}\big(\mathcal{F}^{-1}(M \odot y)\big)$.

The variable-density mask [1] fully samples a central calibration band of n_c low-frequency phase-encode lines and randomly sub-samples higher frequencies with probability $P(k_y) \propto \big(1 - (|k_y|/k_y^{\max})^2\big)^2$, exploiting the energy concentration of natural images at low spatial frequencies. The under-sampling ratio $R = \|M\|_0/(n_r n_c)$ controls the acceleration factor $1/R$.

2.2 CT

For parallel-beam geometry, the forward operator is the 2-D Radon transform $\mathcal{R}$. With projection angles $\theta_j \in [0, \pi)$ and detector coordinate s,

$$A_{\mathrm{CT}}(x)(\theta_j, s) = \mathcal{R}x(\theta_j, s) = \int_{-\infty}^{\infty} x(s\cos\theta_j - t\sin\theta_j,\ s\sin\theta_j + t\cos\theta_j)\, dt. \tag{2}$$

The adjoint $A_{\mathrm{CT}}^* = \mathcal{R}^*$ is the unfiltered back-projection (not the FBP reconstruction, which applies a ramp filter). Sparse-view CT uses $N_\theta \ll N_r$ projection angles, making the problem under-determined. We normalise the operator pair by $1/\sqrt{\pi N_\theta}$ so that $\|A\| \approx 1$, matching the MRI scaling and making regularisation parameters transferable across modalities.

3 Compressed Sensing Principles

Compressed sensing [3, 4] guarantees exact or near-exact recovery of a sparse signal from under-sampled linear measurements under two conditions:

(1) **Sparsity.** The image x admits a sparse representation in a transform domain Ψ: $x = \Psi^* c$, where c has few non-zero entries.

(2) **Incoherence.** The sensing basis (Fourier for MRI; pixel basis for CT) and the sparsity basis (wavelet; finite differences) are sufficiently incoherent.

Under these conditions, the ℓ_1-regularised problem

$$\min_x \frac{1}{2}\|Ax - y\|_2^2 + \lambda \|\Psi x\|_1 \tag{3}$$

recovers the true image with high probability when $M \gtrsim s \log(N/s)$ measurements are acquired, where s is the sparsity level.

4 Variational Reconstruction Methods

4.1 Tikhonov Regularisation (ℓ_2)

The simplest convex regulariser penalises the ℓ_2 energy:

$$\min_x \frac{1}{2}\|Ax - y\|_2^2 + \frac{\lambda}{2}\|x\|_2^2. \tag{4}$$

The solution satisfies the normal equation $(A^*A + \lambda I)x = A^*y$, a symmetric positive-definite linear system solved by the conjugate gradient method. Tikhonov regularisation produces smooth reconstructions; it does not promote sparsity or preserve edges.

4.2 Total Variation (TV)

The discrete isotropic total variation is

$$\mathrm{TV}(x) = \|\nabla x\|_{2,1} = \sum_{i,j} \sqrt{(\nabla_1 x)_{i,j}^2 + (\nabla_2 x)_{i,j}^2}, \tag{5}$$

where ∇_1, ∇_2 are forward finite-difference operators. TV penalises the ℓ_1 norm of the gradient magnitude, promoting piecewise-constant reconstructions with sharp edges.

The reconstruction problem $\min_x \frac{1}{2}\|Ax - y\|_2^2 + \alpha\,\mathrm{TV}(x)$ is solved via the Chambolle-Pock primal-dual algorithm [5]. Defining the saddle-point formulation

$$\min_x \max_{p:\, \|p\|_{2,\infty} \le \alpha} \frac{1}{2}\|Ax - y\|_2^2 + \langle \nabla x, p \rangle,$$

the algorithm alternates dual ascent (pointwise ℓ_2-ball projection), primal descent with the adjoint of $[A;\ \nabla]$, and over-relaxation. The step sizes $\tau = \sigma = 0.99/\sqrt{\|A\|^2 + 8}$ guarantee convergence via $\tau\sigma\|K\|^2 < 1$.

4.3 Wavelet-Domain ℓ_1 Minimisation (FISTA)

Let W be a sparsifying 2-D discrete wavelet transform (e.g., Haar) and W^* its adjoint. The CS objective (3) with $\Psi = W$ is a composite convex problem solved by FISTA [6] with optimal $O(1/k^2)$ convergence:

$$\tilde{x}^k = z^k - \tfrac{1}{L}A^*(Az^k - y), \tag{6}$$

$$x^{k+1} = W^*\big(S_{\lambda/L}(W\tilde{x}^k)\big), \tag{7}$$

$$t_{k+1} = \tfrac{1}{2}\big(1 + \sqrt{1 + 4t_k^2}\big), \qquad z^{k+1} = x^{k+1} + \tfrac{t_k - 1}{t_{k+1}}(x^{k+1} - x^k). \tag{8}$$

Here $L = \|A\|^2$ (estimated by power iteration when not known analytically) and $S_\tau(c) = \mathrm{sign}(c)\max(|c| - \tau, 0)$ is element-wise soft-thresholding.

The Haar wavelet is used as the default sparsifying transform because its piecewise-constant basis functions match the structure of the standard Shepp-Logan phantom, yielding substantially sparser coefficient vectors than smoother wavelets (e.g., Daubechies-4).

5 Package Structure and Availability

The package is organised into flat modules:

Module	Description
`medimg.forward`	MRI (Fourier + mask) and CT (Radon) operators
`medimg.reconstruct`	Tikhonov (CG), TV (Chambolle-Pock), CS (FISTA)
`medimg.transforms`	Wavelet analysis/synthesis, soft-thresholding
`medimg.phantoms`	Shepp-Logan, custom circle phantoms
`medimg.metrics`	PSNR, SSIM (Wang et al., 2004)
`medimg.plotting`	Side-by-side comparisons, error maps

5.1 Installation

```
git clone https://github.com/abdgafartunde/medical-imaging
cd medical-imaging
pip install -e ".[dev]"
```

Dependencies: NumPy (≥ 1.24), SciPy (≥ 1.10), Matplotlib (≥ 3.7), scikit-image (≥ 0.20), PyWavelets (≥ 1.4).

5.2 Quick Start

```
from medimg.phantoms import shepp_logan
from medimg.forward import generate_cartesian_mask, mri_forward, mri_adjoint
from medimg.reconstruct import cs_fista
from medimg.transforms import wavelet_forward, wavelet_adjoint

phantom = shepp_logan((256, 256))
mask    = generate_cartesian_mask(phantom.shape, sampling_rate=0.25,
                                  calib_lines=24, rng=42)
A  = lambda x: mri_forward(x, mask)
At = lambda y: mri_adjoint(y, mask)
y  = mri_forward(phantom, mask, noise_std=0.001, rng=42)
```

```
W  = lambda x: wavelet_forward(x, wavelet='haar', level=4)
Wt = lambda c: wavelet_adjoint(c, wavelet='haar')
x_cs = cs_fista(A, At, y, W, Wt, lambd=1e-4, shape=phantom.shape)
```

5.3 Examples and Tests

```
python examples/01_mri_cs.py        # CS-MRI demo
python examples/02_ct_sparse_view.py # sparse-view CT demo
pytest tests/ -v                     # 41 unit tests
```

6 Image Quality Metrics

PSNR (Peak Signal-to-Noise Ratio, in dB):

$$\mathrm{PSNR}(x, x_{\mathrm{ref}}) = 20 \log_{10}(\mathrm{range}) - 10 \log_{10}\Big(\frac{1}{N} \|x - x_{\mathrm{ref}}\|_2^2\Big).$$

SSIM (Structural Similarity) [7]:

$$\mathrm{SSIM}(x, x_{\mathrm{ref}}) = \frac{(2\mu_x \mu_{\mathrm{ref}} + C_1)(2\sigma_{x,\mathrm{ref}} + C_2)}{(\mu_x^2 + \mu_{\mathrm{ref}}^2 + C_1)(\sigma_x^2 + \sigma_{\mathrm{ref}}^2 + C_2)},$$

where μ, σ are local means and standard deviations over 11×11 Gaussian windows, and C_1, C_2 are stabilisation constants.

References

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